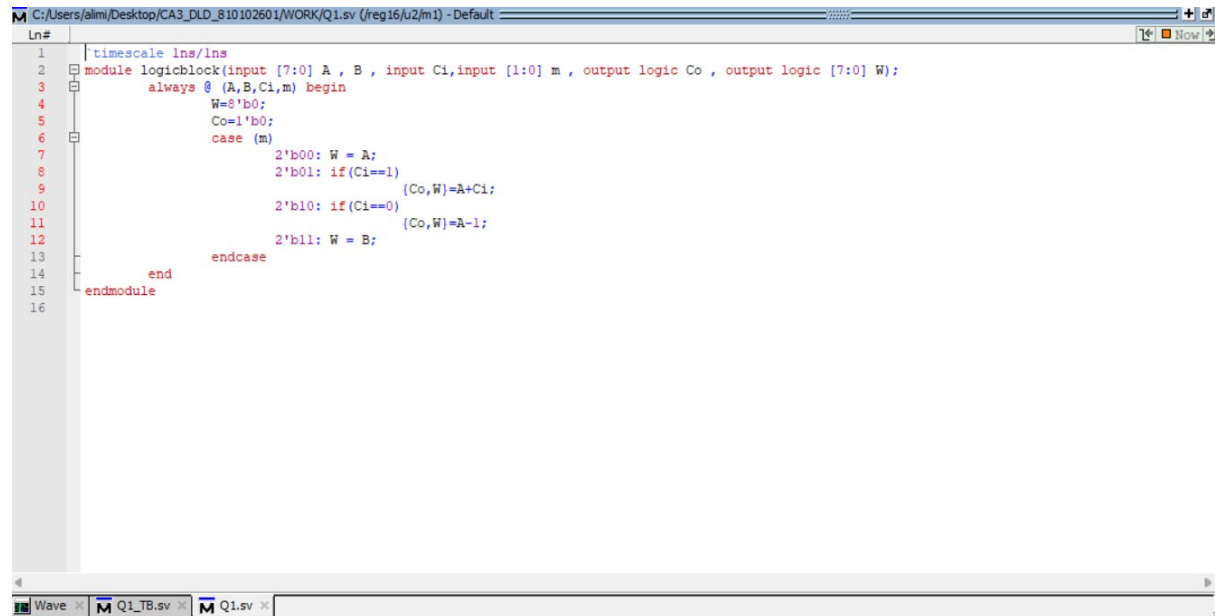


DLD CA 3, Full Name: Amirmohammad Mirhosseini, **Logic No. :** 92

SID: 810102601

Q1)

Verilog Code:



```
1 | timescale 1ns/1ns
2 | module logicblock(input [7:0] A , B , input Ci,input [1:0] m , output logic Co , output logic [7:0] W);
3 |     always @ (A,B,Ci,m) begin
4 |         W=8'b0;
5 |         Co=1'b0;
6 |         case (m)
7 |             2'b00: W = A;
8 |             2'b01: if (Ci==1) (Co,W)=A+Ci;
9 |             2'b10: if (Ci==0) (Co,W)=A-1;
10 |             2'b11: W = B;
11 |         endcase
12 |     end
13 | endmodule
14 |
15 |
16 |
```

Test bench:

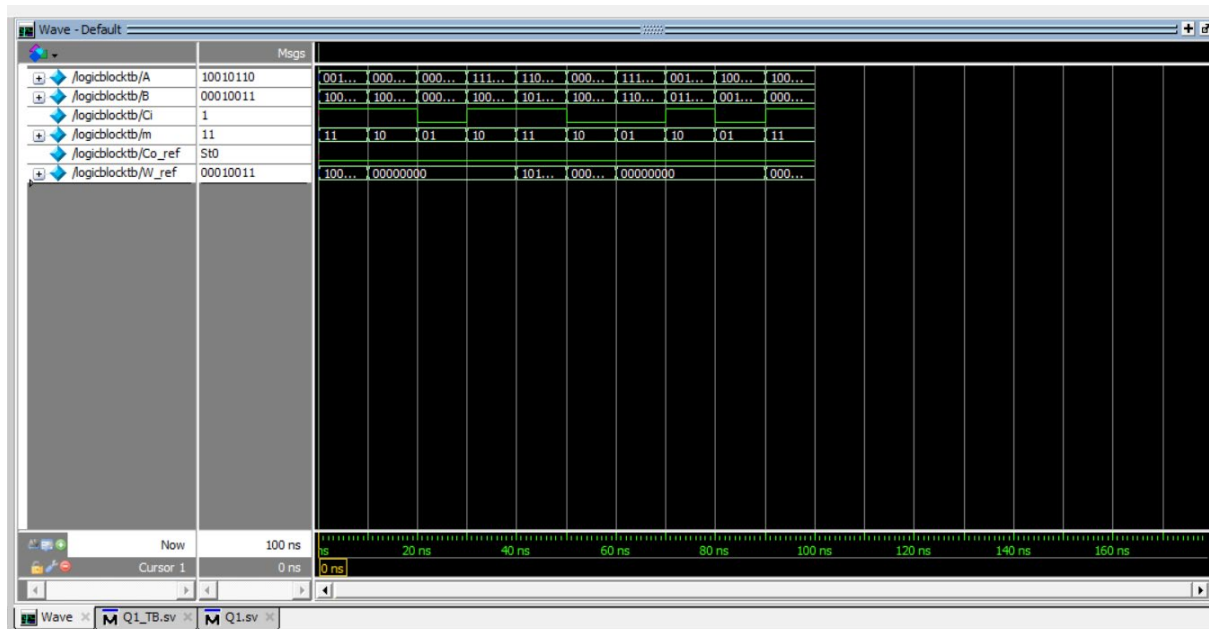
```

Ln#
1 | timescale 1ns/1ns
2 |
3 | module logicblocktb;
4 |
5 |     logic [7:0] A, B;
6 |     logic Ci;
7 |     logic [1:0] m;
8 |
9 |
10 |     wire Co_ref;
11 |     wire [7:0] W_ref;
12 |
13 |     logicblock uut_ref (
14 |         .A(A), .B(B), .Ci(Ci), .m(m), .Co(Co_ref), .W(W_ref)
15 |     );
16 |
17 |
18 |
19 |
20 |     initial begin
21 |         repeat (10) begin
22 |             A = $random;
23 |             B = $random;
24 |             Ci = $random;
25 |             m = $random % 4;
26 |             #10;
27 |         end
28 |         $finish;
29 |     end
30 |
31 | endmodule

```

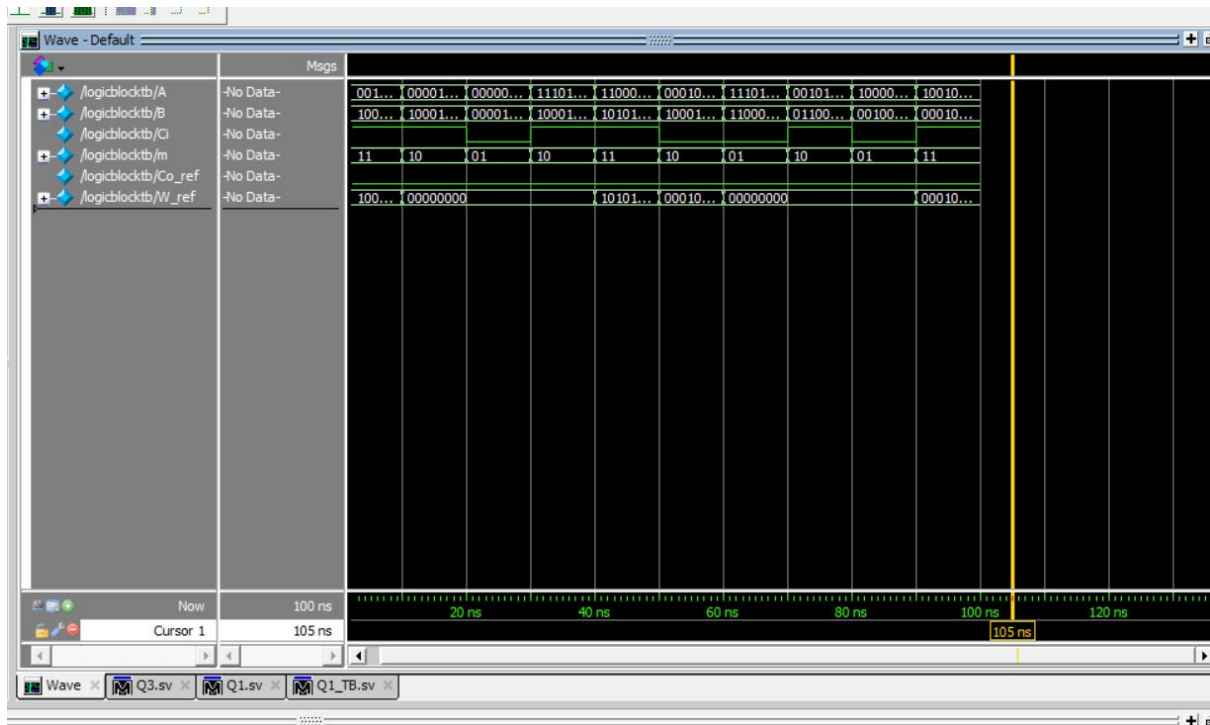
Wave x Q1_TB.sv x Q1.sv x

Waveform:



Q2)

Testing the test bench before using yosys to see it works properly.



As you see, it works pretty well.

After using yosys and saving synthesised file as Q2_SYNTH.sv we will have the results below:

Testbench:

```
#
`timescale 1ns/1ns

module logicblocktb;

    logic [7:0] AA, BB;
    logic Cii;
    logic [1:0] mm;

    wire Co_syn, Co_ref;
    wire [7:0] W_syn, W_ref;

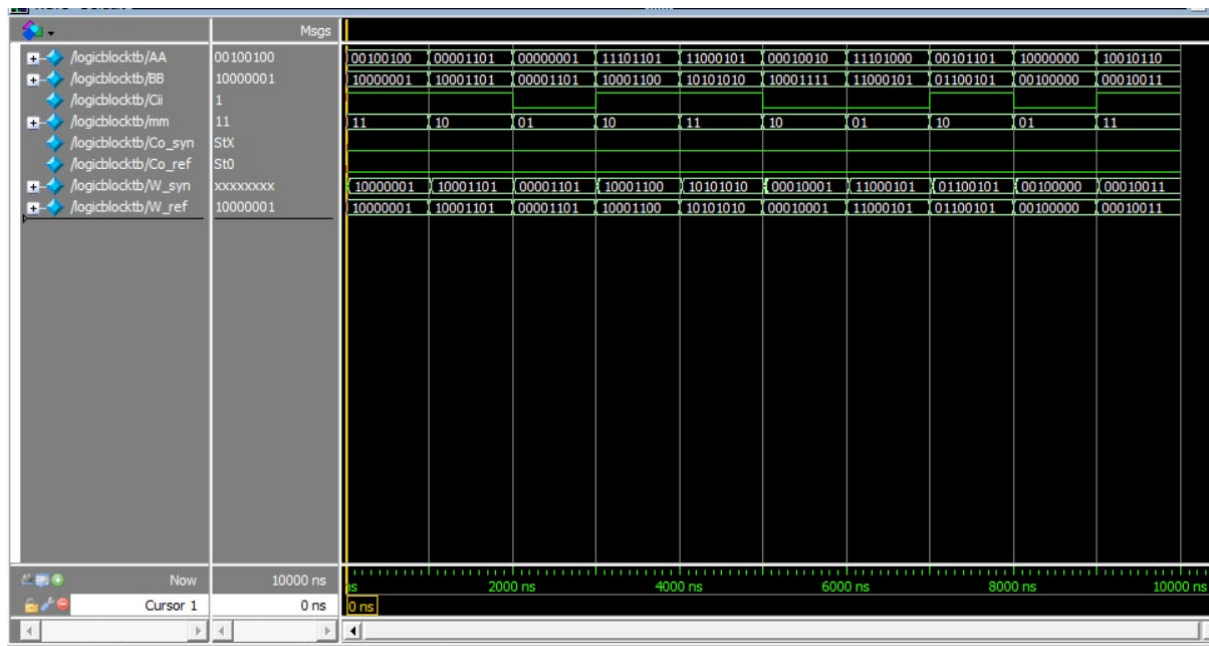
    logicblock uut_ref (
        AA, BB, Cii, mm, Co_ref, W_ref
    );

    logicblock1 uut_syn (
        AA, BB, Cii, mm, Co_syn, W_syn
    );

    initial begin
        repeat (10) begin
            AA = $random;
            BB = $random;
            Cii = $random;
            mm = $random % 4;
            #1000;
        end
        $finish;
    end
end
```

As you can see, we increased the delay between each repeat to stop the possible problems.

Waveforms:



As you can see, it works properly.

Q3)

Verilog Code:

```

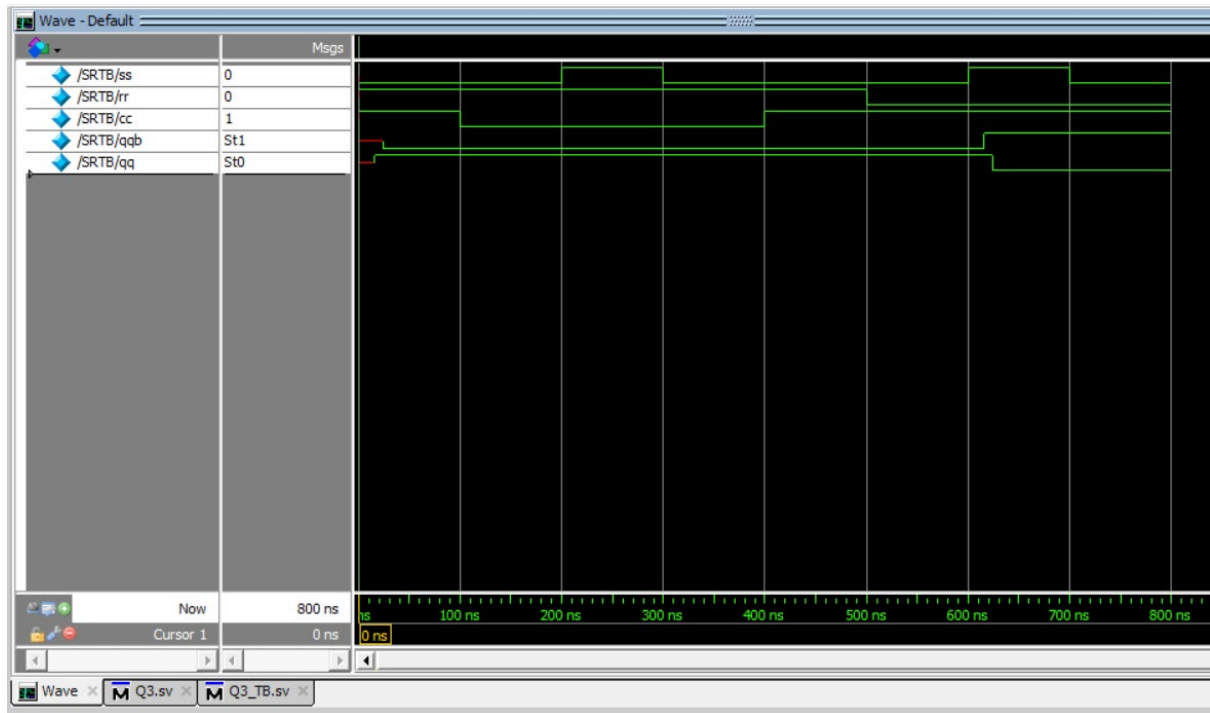
1  `timescale 1ns/1ns
2  module SRlatch(input S,R,clk, output q , output q_bar);
3      wire j , k;
4
5      nand #8 n1(j,clk,S) , n2(k,clk,R) , n3(q,k,q_bar) , n4(q_bar, q,j);
6
7  endmodule
8

```

Test bench:

```
C:/Users/alimi/Desktop/CA3_DLD_810102601/WORK/Q3_TB.sv - Default
Ln#
1  `timescale 1ns/1ns
2  module SRTB();
3      logic ss,rr ,cc;
4      wire qqb;
5      wire qq;
6      //assign qq=0;
7      SRlatch uut(ss,rr,cc,qq,qqb);
8      initial begin
9          rr=1;
10         cc=1;
11         ss=0;
12         #100 cc=0;
13         #100 ss=1;
14         #100 ss=0;
15         #100 cc=1;
16         #100 rr=0;
17         #100 ss=1;
18         #100 ss=0;
19         #100 $stop;
20     end
21 endmodule
22
23
```

Waveform: 



Q4)

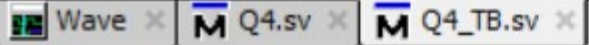
Verilog Code:

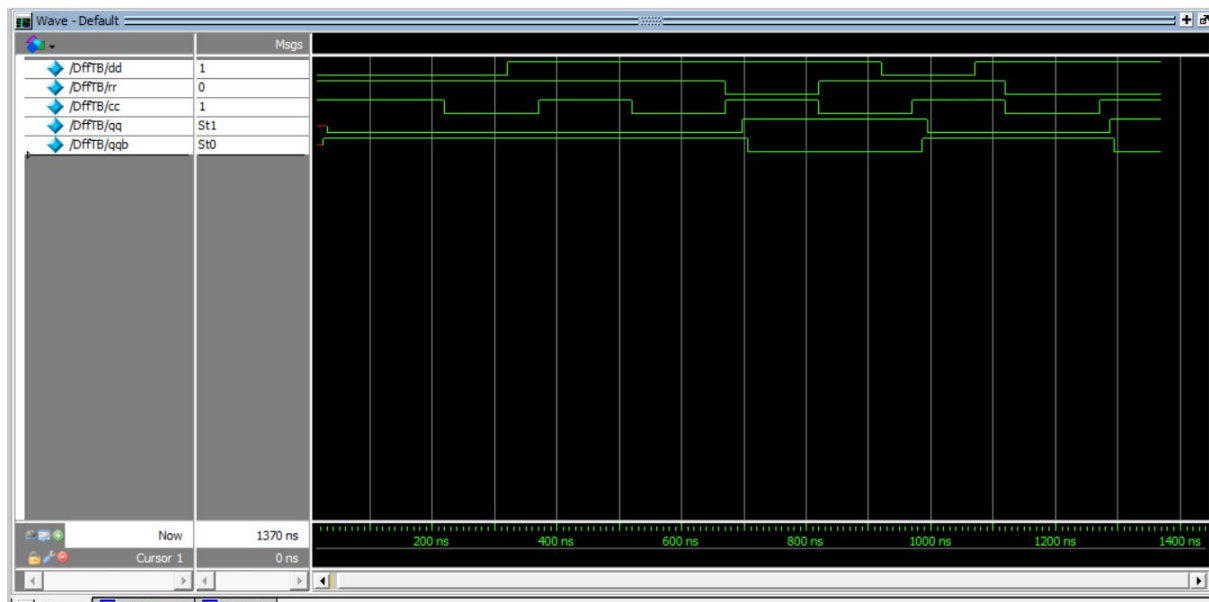
```
C:/Users/alimi/Desktop/CA3_DLD_810102601/WORK/Q4.sv - Default
Ln#
1  `timescale 1ns/1ns
2  module Dff ( input D, r , clk , output Q , q_bar);
3      wire notD, notR, S, Rl;
4      not #6 (notD, D);
5      not #6 (notR, r);
6
7      nand #8 (S, notR, D);
8      nor #12 (Rl, r, notD);
9      SRlatch srl(S, Rl, clk, Q, q_bar);
10
11  endmodule
12
13
```

Wave x Q4.sv x Q4_TB.sv x

Test bench:

```
C:/Users/alimi/Desktop/CA3_DLD_810102601/WORK/Q4_TB.sv - Default
Ln#
1  `timescale 1ns/1ns
2  module DffTB();
3      logic dd, rr, cc;
4      wire qq, qqb;
5
6      Dff uut(dd, rr, cc, qq, qqb);
7
8      initial begin
9          cc = 1;
10         rr = 1;
11         dd = 0;
12         #50;
13         //forever #50 cc = ~cc;
14
15         //rr = 0;
16         #70;
17         // rr = 1;
18         repeat(8) begin
19             #50 dd = $random % 2;
20             #50 rr = $random % 2;
21             cc = ~cc;
22             #50;
23         end
24         #50;
25         $stop;
26     end
27 endmodule
28
29
30
```

Waveform: 



Q5)

Verilog Code:

```

C:/Users/alimi/Desktop/CA3_DLD_810102601/WORK/Q5.sv - Default
Ln#
1  module register8(input [7:0] pin , input clk , r , output logic [7:0] pout , p_b_out);
2      genvar i;
3      generate
4          for(i=0 ; i<8 ; i = i+1) begin
5              Dff tt(pin[i],r,clk,pout[i],p_b_out[i]);
6          end
7      endgenerate
8  endmodule
9

```

The screenshot shows a Verilog code editor window titled "C:/Users/alimi/Desktop/CA3_DLD_810102601/WORK/Q5.sv - Default". The code defines a module named "register8" with inputs "pin" (8-bit), "clk", "r", and outputs "pout" (8-bit logic) and "p_b_out". It uses a "generate" block with a "for" loop to instantiate eight "Dff" (D-type flip-flop) components, each taking a bit of "pin", "r", "clk", "pout", and "p_b_out" as inputs. The code is listed line by line from 1 to 9. The bottom status bar shows three open tabs: "Wave", "Q5_TB.sv", and "Q5.sv".

Test bench:


```

C:/Users/aimi/Desktop/CA3_DLD_810102601/WORK/Q6.sv - Default
Ln#
1 | timescale 1ns/1ns
2 | module latchCounter(input [7:0] B, input [1:0] m, input Ci, input clk, input rst, output [7:0] Count, output Co);
3 |     wire [7:0] W, Count_bar;
4 |
5 |     logicblock ml(Count , B , Ci , m , Co, W);
6 |     register8 m2(W , clk , rst , Count, Count_bar);
7 | endmodule
8

```

Test bench:

```

Ln#
2 | module latchREGISTERtb();
3 |     reg [7:0] B;
4 |     reg [1:0] m;
5 |     reg Ci , clk , rst;
6 |     wire [7:0] Count;
7 |     wire Co;
8 |     latchCounter uut(B , m , Ci , clk , rst , Count , Co);
9 |     initial begin
10 |         B = 8'b0;
11 |         m = 2'b11;
12 |         clk = 1;
13 |         Ci = 1;
14 |         rst = 0;
15 |         #20 clk = 0 ;
16 |         #20 m = 2'b01;
17 |         #60 ;
18 |         repeat (60) #20 clk = ~clk;
19 |         #60 rst = 1;
20 |         #30 clk = 1;
21 |         #30 rst = 0;
22 |         repeat (50) #20 clk = ~clk;
23 |         #25 Ci = 0;
24 |         #60 m = 2'b10;
25 |         repeat (30) #20 clk = ~clk;
26 |         #20 clk = 0;
27 |         #60 B = 8'b11111111;
28 |         #20 m = 2'b11;
29 |         #10 clk = 1;
30 |         #20 clk = 0;
31 |         #5 Ci = 1;
32 |         #20 m = 2'b01;

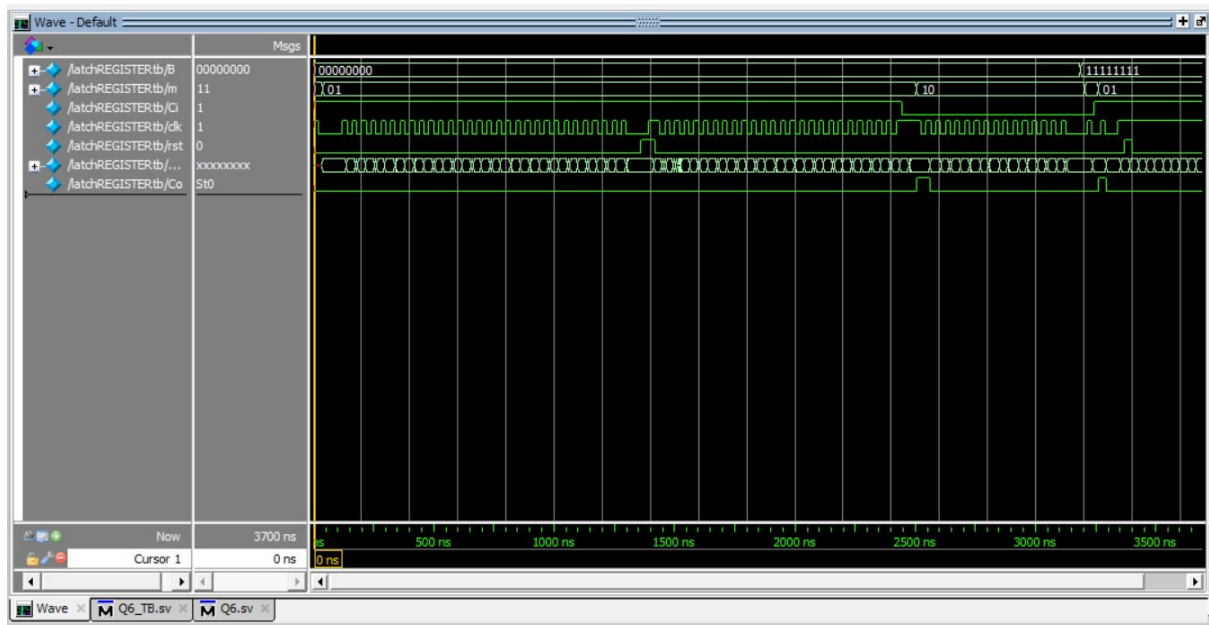
```

```

33 |         #10 clk = 1;
34 |         #20 clk = 0;
35 |
36 |         #50 clk=1;
37 |         #30 rst=1;
38 |         #30 rst=0;
39 |         #300 $stop;
40 |     end
41 | endmodule
42

```

Waveform:



For post-synth

Verilog Code:

```

C:\Users\alim\Desktop\CA3_DLD_810102601\WORK\Q6_SYN.sv - Default
Ln#
1 | timescale 1ns/1ns
2 | module SYNlatchCounter(input [7:0] B, input [1:0] m, input Ci, input clk, input rst, output [7:0] Count, output Co);
3 |     wire [7:0] W, Count_bar;
4 |
5 |     logicblock1 m1(Count, B, Ci, m, Co, W);
6 |     register8 m2(W, clk, rst, Count, Count_bar);
7 | endmodule
8 |

```

Wave x M Q6_TB_SYNTH.sv x M Q6_SYN.sv x

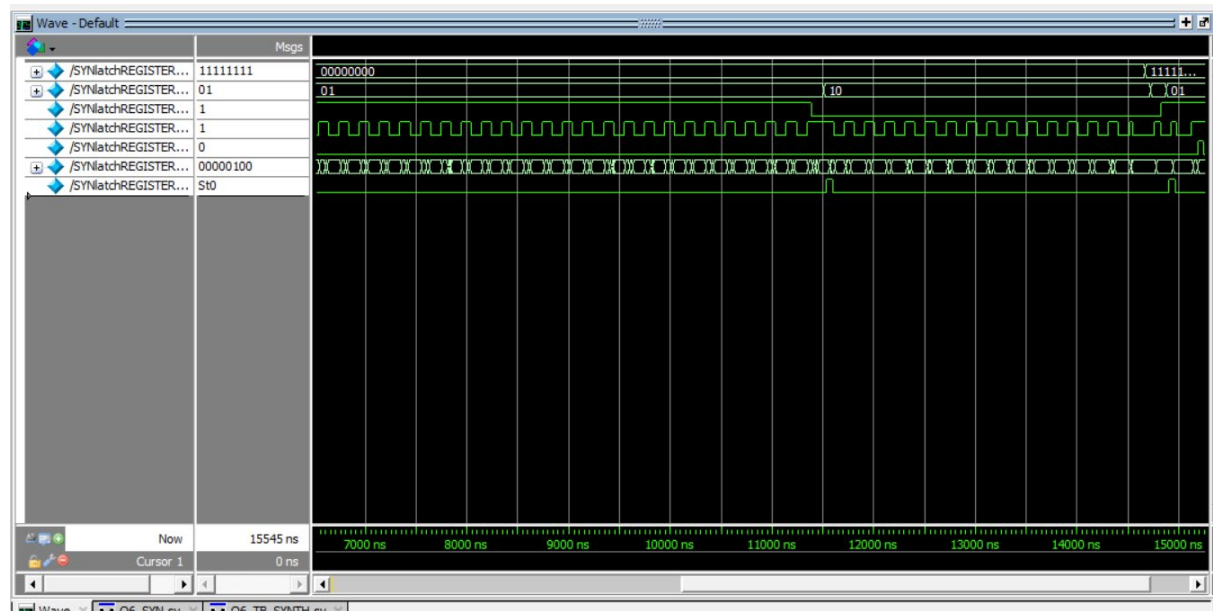
Test bench:

```

C:/Users/alimi/Desktop/CA3_DLD_810102601/WORK/Q6_TB_SYNTH.sv - Default
Ln#
2 module SYNlatchREGISTERtb();
3     reg [7:0] B;
4     reg [1:0] m;
5     reg Ci, clk, rst;
6     wire [7:0] Count;
7     wire Co;
8     SYNlatchCounter uut(B, m, Ci, clk, rst, Count, Co);
9     initial begin
10        B = 8'b0;
11        m = 2'b11;
12        clk = 1;
13        Ci = 1;
14        rst = 0;
15        #20 clk = 0;
16        #20 m = 2'b01;
17        #60;
18        repeat (60) #100 clk = ~clk;
19        #120 rst = 1;
20        #60 clk = 1;
21        #60 rst = 0;
22        repeat (50) #100 clk = ~clk;
23        #50 Ci = 0;
24        #120 m = 2'b10;
25        repeat (30) #100 clk = ~clk;
26        #40 clk = 0;
27        #120 B = 8'b11111111;
28        #50 m = 2'b11;
29        #40 clk = 1;
30        #60 clk = 0;
31        #5 Ci = 1;
32        #50 m = 2'b01;
33
34        #50 clk = 1;
35        #50 clk = 0;
36
37        #150 clk=1;
38        #60 rst=1;
39        #60 rst=0;
40        #300 $stop;
41    end
42 endmodule

```

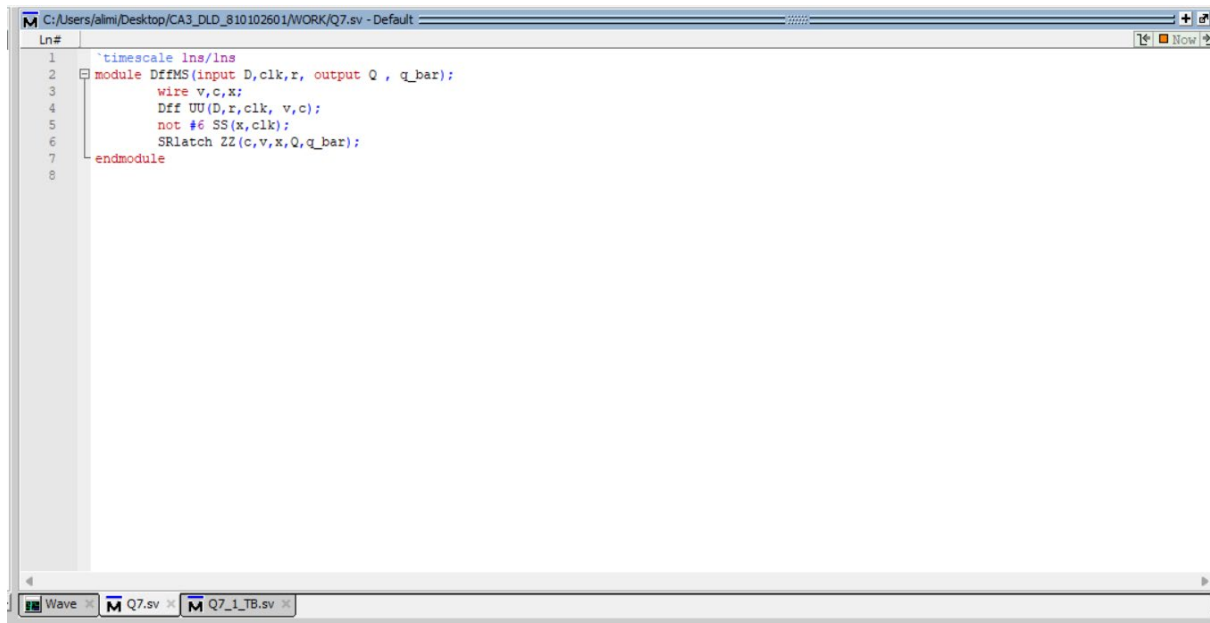
Waveform:



Q7)

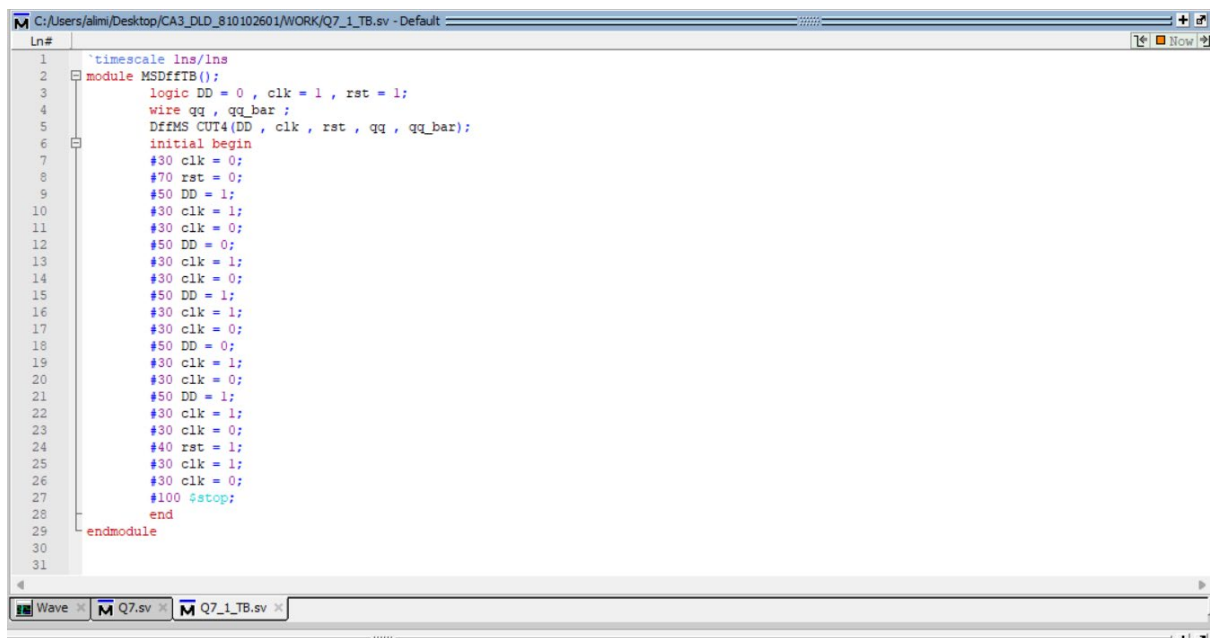
MSDFF:

Verilog Code:



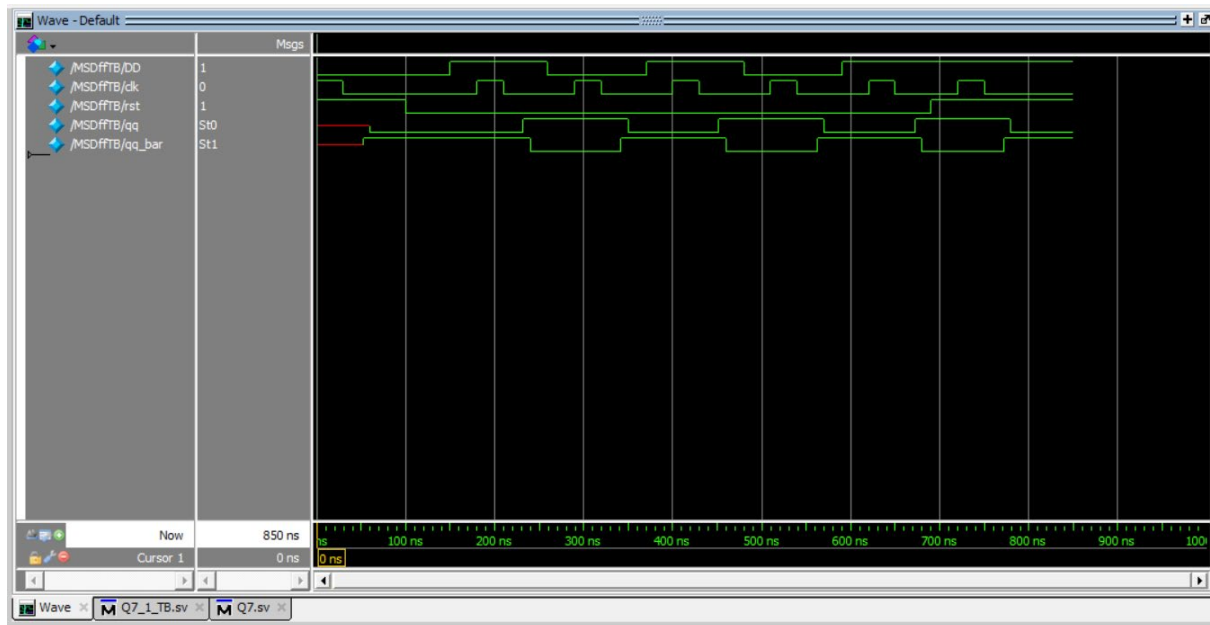
```
1 `timescale 1ns/1ns
2 module DffMS(input D,clk,r, output Q , q_bar);
3     wire v,c,x;
4     Dff UU(D,r,clk, v,c);
5     not #6 SS(x,clk);
6     SRlatch ZZ(c,v,x,Q,q_bar);
7 endmodule
8
```

Test bench:



```
1 `timescale 1ns/1ns
2 module MSDffTB();
3     logic DD = 0 , clk = 1 , rst = 1;
4     wire qq , qq_bar ;
5     DffMS CUT4(DD , clk , rst , qq , qq_bar);
6     initial begin
7         #30 clk = 0;
8         #70 rst = 0;
9         #50 DD = 1;
10        #30 clk = 1;
11        #30 clk = 0;
12        #50 DD = 0;
13        #30 clk = 1;
14        #30 clk = 0;
15        #50 DD = 1;
16        #30 clk = 1;
17        #30 clk = 0;
18        #50 DD = 0;
19        #30 clk = 1;
20        #30 clk = 0;
21        #50 DD = 1;
22        #30 clk = 1;
23        #30 clk = 0;
24        #40 rst = 1;
25        #30 clk = 1;
26        #30 clk = 0;
27        #100 $stop;
28    end
29 endmodule
30
31
```

Waveform:



8 bit register with MSDFF:

Verilog code:

```

1 | timescale 1ns/1ns
2 | module MSRegister8 (input [7:0] pin, input R, clk, output [7:0] pout, pb_out);
3 |
4 |     genvar q;
5 |     generate
6 |         for (q=0; q<8; q=q+1) begin
7 |             DffMS uu(pin[q], clk, R, pout[q], pb_out[q]);
8 |         end
9 |     endgenerate
10 |
11 | endmodule
12 |

```

8 bit register with MSDFF(ALWAYS):

Verilog code:

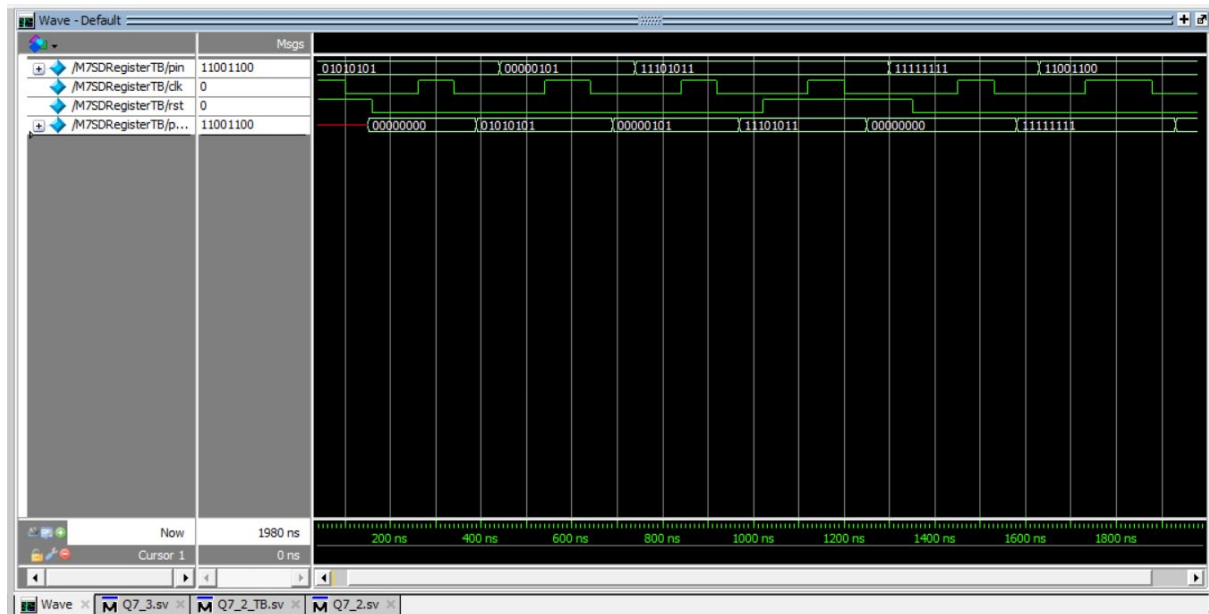
```
1 timescale 1ns/1ns
2 module MSRegister8ALWS(input [7:0] Pi , input clk , R , output reg [7:0] Po);
3     always @(negedge clk) begin
4         if(R)
5             Po <= #50 8'b0;
6         else
7             Po<= #50 Pi;
8         end
9     endmodule
10
```

Waveform viewer tabs: Wave, M Q7_2.sv, M Q7_3.sv, M Q7_2_TB.sv

Testbench:

```
module M7SDRegisterTB();
    logic [7:0] pin ;
    logic clk = 1 ;
    logic rst = 1;
    wire [7:0] pout;
    MSRegister8ALWS uut(pin , clk , rst , pout);
    initial begin
        pin = 8'b01010101;
        #100 clk = 0;
        #60 rst = 0;
        #100 clk = 1;
        #80 clk = 0;
        #100 pin = 8'b00000101;
        #100 clk = 1;
        #100 clk = 0;
        #100 pin = 8'b11101011;
        #100 clk = 1;
        #80 clk = 0;
        #100 rst = 1;
        #100 clk = 1;
        #80 clk = 0;
        #100 pin = 8'b11111111;
        #50 rst = 0;
        #100 clk = 1;
        #80 clk = 0;
        #100 pin = 8'b11001100;
        #100 clk = 1;
        #150 clk = 0;
        #100 $stop;
    end
endmodule
```

Waveform:



Q8)

PRE-SYNTH:

Verilog Code:

```

Ln# | timescale 1ns/1ns
1 | module MSDffREG8(input [7:0] B, input [1:0] m, input Ci, input clk, input rst, output [7:0] Count, output Co);
2 |   wire [7:0] W;
3 |   logicblock m1(Count, B, Ci, m, Co, W);
4 |   MSRegister8ALWS m2(W, clk, rst, Count);
5 | endmodule
6 |
7 |
8 |

```

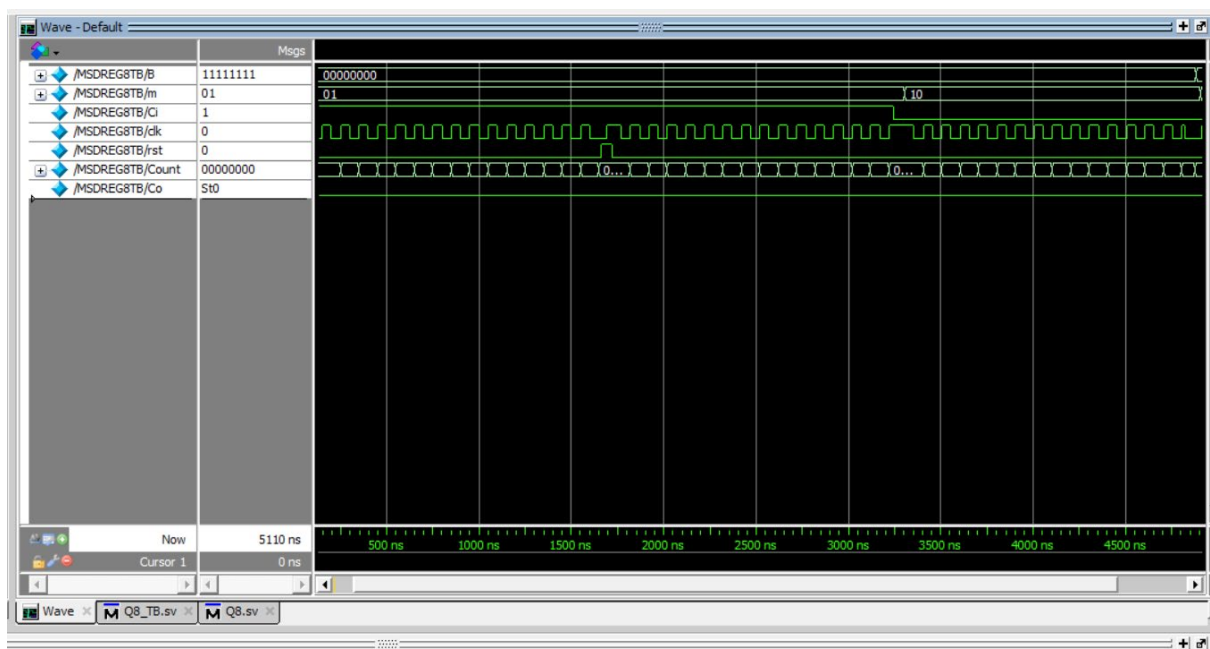
Test bench:

```

2  module MSDREG8TB();
3      logic [7:0] B;
4      logic [1:0] m;
5      logic Ci, clk, rst;
6      wire [7:0] Count;
7      wire Co;
8      MSDffREG8 uut(B, m, Ci, clk, rst, Count, Co);
9      initial begin
10         B = 8'b00000000;
11         m = 2'b11;
12         clk = 1;
13         Ci = 1;
14         rst = 0;
15         #20 clk = 0;
16         #20 m = 2'b01;
17         #60;
18         repeat (30) #50 clk = ~clk;
19         #60 rst = 1;
20         #30 clk = 1;
21         #30 rst = 0;
22         repeat (30) #50 clk = ~clk;
23         #25 Ci = 0;
24         #60 m = 2'b10;
25         repeat (30) #50 clk = ~clk;
26         #20 clk = 0;
27         #60 B = 8'b11111111;
28         #20 m = 2'b11;
29         #10 clk = 1;
30         #50 clk = 0;
31         #5 Ci = 1;
32         #20 m = 2'b01;
33
34         #20 m = 2'b01;
35         #10 clk = 1;
36         #50 clk = 0;
37         #60 $stop;
38     end
39 endmodule

```

Waveform:



POST-SYNTH:

Verilog Code:

```

C:/Users/alini/Desktop/CA3_DLD_810102601/WORK/SynMSDFREG8.sv - Default
Ln#
1 timescale 1ns/1ns
2 module SYNMSDffREG8(input [7:0] B, input [1:0] m, input Ci, input clk, input rst, output [7:0] Count, output Co);
3 wire [7:0] W;
4
5 logicblock1 m1(Count , B , Ci , m , Co, W);
6 MSRegister8ALWS m2(W , clk , rst , Count);
7 endmodule
8

```

Waveform viewer tabs: Wave, SynMSDFREG8.sv

Test bench:

```

Ln#
1 timescale 1ns/1ns
2 module TB SYNMSDffREG8();
3 logic [7:0] B;
4 logic [1:0] m;
5 logic Ci, clk, rst;
6 wire [7:0] Count;
7 wire Co;
8 SYNMSDffREG8 uut(B, m, Ci, clk, rst, Count, Co);
9 initial begin
10 B = 8'b00000000;
11 m = 2'b11;
12 clk = 1;
13 Ci = 1;
14 rst = 0;
15 #120 clk = 0;
16 #120 m = 2'b01;
17 #160;
18 repeat (30) #150 clk = ~clk;
19 #160 rst = 1;
20 #130 clk = 1;
21 #130 rst = 0;
22 repeat (30) #150 clk = ~clk;
23 #125 Ci = 0;
24 #160 m = 2'b10;
25 repeat (30) #150 clk = ~clk;
26 #120 clk = 0;
27 #160 B = 8'b11111111;
28 #120 m = 2'b11;
29 #110 clk = 1;
30 #150 clk = 0;
31 #15 Ci = 1;

```

Waveform viewer tabs: Wave, SynTBMSDFREG8.sv, SynMSDFREG8.sv

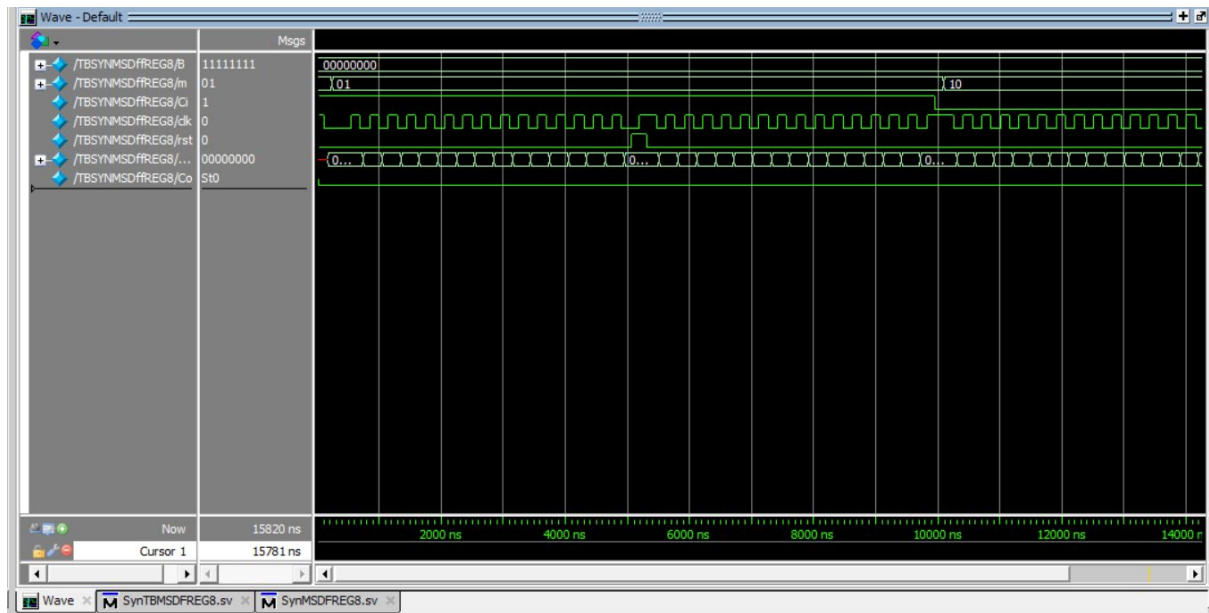
```

30 #150 clk = 0;
31 #15 Ci = 1;
32 #120 m = 2'b01;
33 #110 clk = 1;
34 #150 clk = 0;
35 #160 $stop;
36 end
37 endmodule
38

```

Waveform viewer tabs: Wave, SynTBMSDFREG8.sv, SynMSDFREG8.sv

Waveform:



Q9)

Verilog Code:

```

1  `timescale 1ns/1ns
2  module reg16(input [7:0] B0t7, B8t15, input [1:0] m, input Ci, input clk, rst, output [15:0] Count, output Co);
3      wire co_cin;
4      MSDffREG8 u1(B0t7, m, Ci, clk, rst, Count[7:0], co_cin);
5      MSDffREG8 u2(B8t15, m, co_cin, clk, rst, Count[15:8], Co);
6  endmodule
7

```